

Alexander Reinthal

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I am an independent AI safety researcher studying how models become subtly misaligned during training. I bring 7 years of industry experience across cyber-security, data, and ML engineering.

Research Projects

Testing Scheming Drivers on Instrumental Convergence Evals

May 2026 – June 2026

What happens to instrumental convergence evals when you change the prompt?

- Found that pressure phrases like ‘achieve your goal in the long run’ have opposite effects on misalignment rates in Gemini 3.1 Pro depending on prompt styles (1) [He et al. \(2025\)](#) capability prompt and (2) [Hopman et al. \(2026\)](#) propensity prompt. Pressure phrases increased the misalignment rate for (1) +0.23 [+0.19, +0.28] but decreased for (2) −0.10 [−0.15, −0.05] 95% Newcombe CI.

Detecting Deception in Chinese Models

Jan 2026 – present

Can a lying linear probe flag Chinese models answers on politically sensitive questions?

- We found that Chinese models (Qwen2.5-72B and Qwen3.5-122B-A10B) answer more evasively, state contested information without caveats, use faulty logic and refuse sensitive political questions more compared to western models (Llama3.3-70B and GPT-OSS-120B) but that deception probes (Apollo, 2025) are unreliable at detecting such problematic behavior.
- Extension of ARENA 7.0 weekend hackathon, 1st place. Supported by Bluedot Impact rapid grant.

About Emergent Misalignment

Jan 2026 – present

Explored inoculation prompting and harmful generalization in LLMs.

- Found that emergent misalignment (EM) rate is negatively correlated on the capability benchmarks GSM8K (pearson −0.98) and MMLU (pearson −0.94) when induced by training on increasing percentages of bad medical advice.
- Extension of ARENA 7.0 capstone, 3rd place.

Detecting Piecewise Cyber Espionage in Model APIs

Nov 2025 – Nov 2025

Apart Research sprint, 4th of 641 submissions — proof of concept for detecting malicious activity across multiple agent sessions.

- Individual requests can look benign in isolation (100% to 23% detection by LlamaGuard 3 on IP scans) — but can be detected by modelling activity across user sessions.
- Co-authored with Arthur Colle, David Williams-King, Yingquan Li, and Lihn Le.

Experience

AI Safety Researcher

May 2026 – present

Independent

- Researching how subtle misalignment develops during training. Inspired by ARENA seminars on RL, inoculation prompting, and natural emergent misalignment, and by my ARENA capstone. Supported by Coefficient Giving’s Career Transition grant for reducing global catastrophic risks.

Mentor

Mar 2026 – present

Bluedot Impact

- Part-time facilitator and mentor for the Technical AI Safety course and project.

Alum

Jan 2026 – Feb 2026

ARENA 7.0

- Five-week intensive curriculum on transformer internals, mechanistic interpretability, RLHF, evals, JAX, and the Inspect framework.

Data Platform Engineer / Tech Lead / Data Scientist

Gothenburg, Sweden

Knowit Solutions Cocreate

Feb 2022 – present

- Implemented an OAuth-authenticated MCP gateway to secure agentic workflows and pioneered LLM evals at the company.

Security Analyst

Gothenburg, Sweden

NTT Security

Apr 2019 – Jan 2022

- Built ML classifiers and anomaly-detection pipelines for malicious network traffic across high-volume customer environments using Python and distributed processing.

Python Software Engineer

Gothenburg, Sweden

Ericsson

June 2017 – Oct 2018

- Led a team building a log-parsing and analytics tool that grew into a dedicated Ericsson engineering team.

Publications

Data Modeling for Predicting Software Exploits

Nov 2018

Peer-reviewed paper for predicting which CVEs become exploited in the wild. Adopted by Recorded Future's Vulnerability Intelligence product (threat-intelligence company). Customers report 86% reduction in unplanned downtime, 11 hours/week saved on triage, and 73% increase in threat visibility.

Reinthal, A., Filippakis, E., Almgren, M.

[10.1007/978-3-030-03638-6_21](https://doi.org/10.1007/978-3-030-03638-6_21) (Springer LNCS: Nordic Conference on Secure IT Systems)

Open Source and Personal Projects

[Casually Jailbreaking Gemini 2.5 Flash](#)

2026 – 2026

Jailbroke Gemini 2.5 Flash with a few-shot in-context continuation attack, eliciting harmful recommendations.

- Practical demonstration of adversarial-robustness gaps in a production frontier model — directly relevant to the team's automated red-teaming and multi-turn attack work.

Open Source Contributions

github.com/reinthal

- [Inspect AI PR #3987](#) Fixed Gemini Pro 3.1+Openrouter+Reasoning issue
- [Dagster PR #24188](#) (ML pipeline orchestration) and [DLT Hub PR #594](#) (ingestion reliability)

Skills

Machine Learning & AI: PyTorch, JAX (learned during ARENA 7.0), Einops, XGBoost, SciKit Learn, Jupyter; LLM evals & observability (Inspect, LangChain, Langfuse, Arize Phoenix); linear probes, SFT, automated red-teaming

Programming Languages: Python, JavaScript, SQL, Rust, C/C++, R, Nix, Terraform

Platforms & Tools: Modal, Runpod, Openweights, Weights & Biases, Spark, Docker, Kubernetes, AWS, Azure

Certifications: Bluedot Impact — Technical AI Safety (2026-03); AGI Strategy (2025-10)

Education

Chalmers University of Technology

MS in Engineering Physics

Gothenburg, Sweden

Sept 2016 – June 2018

- Coursework: Statistical Physics, Neural Networks and Machine Learning

Gothenburg University

BS in Computer Science

Gothenburg, Sweden

Sept 2013 – June 2016

- Coursework: Algorithms, Testing/Debugging/Verification, Theoretical Computer Science, Operating Systems, Cryptography, Cyber Security